Qualification Pack

Solar Photovoltaic Technician

QP Code: SGJ/Q4004

Version: 1.0

NSQF Level: 4

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
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Contents

SGJ/Q4004: Solar Photovoltaic Technician

Brief Job Description

Solar PV Technician shall perform testing, installation and facility integration of various parts, repairs, troubleshooting, upkeep and maintenance of electrical control systems, protection systems, and other auxiliary equipment and associated tools in Solar PV System. The job holder will be responsible for the complete installation, maintenance, electric wiring Solar Pv system for Production of Solar Power. This role works closely with the power supply project, testing, plant engineering, process operation, control & operation all types of solar Pv Plants.

Personal Attributes

Solar PV Technician is responsible for installation of Solar Pv project setup along with thorough monitoring of the project performance and ensuring proper maintenance of each and every equipment to ensure optimum service delivery.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. SGJ/N4023: Introduce Solar PV Technology
2. SGJ/N4024: Perform DC electrical work of Solar Module components
3. SGJ/N4025: Identify and Explain Solar PV System Configuration and identify components
4. SGJ/N4026: Perform site assessment and interpret the layout
5. SGJ/N4027: Install Solar PV DC Standalone system (St. Light)
6. SGJ/N4028: Install Solar Off Grid PV System
7. SGJ/N4029: Install Small solar rooftop grid tied plant for residential home
8. SGJ/N4030: Introduce tools and tackles used for Solar Installation
9. SGJ/N4031: Maintain Health and safety measures for installing and operating solar DC system
10. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Green Jobs

Qualification Pack

Sub-Sector	Renewable Energy
Occupation	Technician
Country	India
NSQF Level	4
Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7421.1402
Minimum Educational Qualification & Experience	11th grade pass (and pursuing continuous schooling) with NA of experience OR Pursuing 2nd year of 2-year diploma after 12th (or Completed 2nd year of 3-year diploma (after 10th)) with NA of experience OR 10th grade pass (with two years of any combination of NTC/NAC/CITS or equivalent) with NA of experience OR 11th grade pass with 1 Year of experience relevant experience OR 10th grade pass with 2 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (Level 3.5) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (Level 3) with 3-5 Years of experience with 3.5 years of relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0

Reference code on NQR	QG-04-ES-00910-2023-V1-SCGJ
NQR Version	1

Remarks:

Total 390 Hours i.e. 13 Credits (Including 330 Hours of Mandatory NOS with 60 hours of Employability Skills)

SGJ/N4023: Introduce Solar PV Technology

Description

This unit is about introduce Solar PV Technology

Scope

The scope covers the following :

- Introduce Solar PV Technology
- basic structure of Solar cell and solar module
- solar cell schematics and identify functions of the key components

Elements and Performance Criteria

Introduce Solar PV Technology

To be competent, the user/individual on the job must be able to:

PC1. identify all key components of the solar cell and solar module and illustrate the schematic of solar cell and solar module

PC2. discuss functions of the key components of solar cell and illustrate key components of the solar cell

PC3. explain type of solar cell and solar module

PC4. explain key material used in fabrication of solar cell and solar module and illustrate working principle of silicon solar cell

PC5. explain working principle of solar cell and solar module and demonstrate different type of solar cell and solar module

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy.

KU2. Company's work safety policy

KU3. Company's Customer Support Policy.

KU4. Company's documentation policy

KU5. Obtain authorization from specified field safety officer and supervisor

KU6. Company's reporting structure and Organization culture

KU7. Company's different department and concerned authority

KU8. Relevant Personal protective equipment's required for installation

KU9. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU10. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU11. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

KU12. Knowhow of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role

GS2. Read English and/or vernacular language

GS3. Read and understand manuals, health and safety instructions, memos, other company documents

GS4. Ability to read from different sources- books screens in machines and signage

GS5. Understand the various color codes, as per standard electrical, mechanical

GS6. Express statements or information clearly so that others can hear and understand

GS7. Participate in and understand the main points of simple discussions

GS8. Respond appropriately to any queries

GS9. Communicate with supervisor

GS10. Follow organization rule-based decision making process

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduce Solar PV Technology	**26**	**24**	**-**	**-**
PC1. identify all key components of the solar cell and solar module and illustrate the schematic of solar cell and solar module	5	6	-	-
PC2. discuss functions of the key components of solar cell and illustrate key components of the solar cell	5	6	-	-
PC3. explain type of solar cell and solar module	6	-	-	-
PC4. explain key material used in fabrication of solar cell and solar module and illustrate working principle of silicon solar cell	5	6	-	-
PC5. explain working principle of solar cell and solar module and demonstrate different type of solar cell and solar module	5	6	-	-
NOS Total	**26**	**24**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4023
NOS Name	Introduce Solar PV Technology
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4024: Perform DC electrical work of Solar Module components

Description

This unit is about perform DC electrical work of Solar Module components

Scope

The scope covers the following :

- Solar Interconnection procedure
- Open circuit voltage and short circuit current
- series and parallel connection

Elements and Performance Criteria

Perform DC electrical work of Solar Module components

To be competent, the user/individual on the job must be able to:

PC1. discuss the interconnection procedure

PC2. discuss the series and parallel connection of solar cell and solar module and demonstrate the series and parallel connection of solar module

PC3. explain the string and array definition of solar module

PC4. discuss the string voltage and current

PC5. discuss the Open circuit voltage and short circuit current and measure the open circuit voltage and short circuit current of solar module

PC6. discuss the Vmp (voltage at maximum power) and Imp(Current at maximum power) of Solar module and measure the Vmp and Imp of solar module

PC7. discuss the standard test conditions of Solar module

PC8. discuss the different power rating of solar module and its importance and demonstrate different type of solar cell and solar module

PC9. discuss the solar module specification and its role in power generation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's Customer Support Policy

KU4. Company's documentation policy

KU5. Obtain authorization from specified field safety officer and supervisor

KU6. Company's reporting structure and Organization culture

KU7. Company's different department and concerned authority

KU8. Relevant Personal protective equipment's required for installation

KU9. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU10. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU11. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

KU12. Knowhow of tools & tackles required to carry out the work.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role

GS2. Read English and/or vernacular language

GS3. Read and understand manuals, health and safety instructions, memos, other company documents

GS4. Ability to read from different sources- books screens in machines and signage

GS5. Understand the various color codes, as per standard electrical, mechanical

GS6. Express statements or information clearly so that others can hear and understand

GS7. Participate in and understand the main points of simple discussions

GS8. Respond appropriately to any queries

GS9. Communicate with supervisor

GS10. Follow organization rule-based decision making process

GS11. Take decision with systematic course of actions and/or response

GS12. Planning and organization of work to meet deadlines

GS13. Work constructively and collaboratively with others

GS14. Follow code of conduct

GS15. Manage relationships with customers with intent on satisfying its requirements for service delivery

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform DC electrical work of Solar Module components	**30**	**20**	**-**	**-**
PC1. discuss the interconnection procedure	3	-	-	-
PC2. discuss the series and parallel connection of solar cell and solar module and demonstrate the series and parallel connection of solar module	3	5	-	-
PC3. explain the string and array definition of solar module	3	-	-	-
PC4. discuss the string voltage and current	3	-	-	-
PC5. discuss the Open circuit voltage and short circuit current and measure the open circuit voltage and short circuit current of solar module	3	5	-	-
PC6. discuss the Vmp (voltage at maximum power) and Imp(Current at maximum power) of Solar module and measure the Vmp and Imp of solar module	3	5	-	-
PC7. discuss the standard test conditions of Solar module	4	-	-	-
PC8. discuss the different power rating of solar module and its importance and demonstrate different type of solar cell and solar module	3	5	-	-
PC9. discuss the solar module specification and its role in power generation	5	-	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4024
NOS Name	Perform DC electrical work of Solar Module components
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4025: Identify and Explain Solar PV System Configuration and identify components

Description

This unit is about identify and explain Solar PV System Configuration and identify components

Scope

The scope covers the following :

- Solar PV System types and its component
- electrical assembly of various solar PV system type
- the schematics and identify functions of the key components

Elements and Performance Criteria

Identify and Explain Solar PV System Configuration and components

To be competent, the user/individual on the job must be able to:

PC1. discuss the home lighting system configuration and its components
PC2. discuss the distributed solar PV system and its components
PC3. discuss the charge controller function and its operation
PC4. discuss the configuration of charge controller for various solar PV application
PC5. discuss the type of charge controller and advantage and disadvantage
PC6. discuss the solar PV mini and micro grid system as distributed solar power generation
PC7. discuss the small solar pv application for lighting and other application
PC8. discuss the configuration and components of standalone Offgrid solar PV system
PC9. demonstrate the series and parallel connection of solar module
PC10. show how to measure the open circuit voltage (Voc) and short circuit current (Isc) of solar module
PC11. show how to measure the Vmp and Imp of solar module
PC12. demonstrate different type of solar cell and solar module

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy
KU2. Company's work safety policy
KU3. Company's documentation policy
KU4. Obtain authorization from specified field safety officer and supervisor
KU5. Company's reporting structure and Organization culture
KU6. Company's different department and concerned authority

KU7. Relevant Personal protective equipment's required for installation
KU8. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India
KU9. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant
KU10. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant
KU11. Know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role
GS2. Read English and/or vernacular language
GS3. Read and understand manuals, health and safety instructions, memos, other company documents
GS4. Ability to read from different sources- books screens in machines and signage
GS5. Understand the various color codes, as per standard electrical, mechanical
GS6. Express statements or information clearly so that others can hear and understand
GS7. Participate in and understand the main points of simple discussions
GS8. Respond appropriately to any queries.
GS9. Communicate with supervisor
GS10. Follow organization rule-based decision making process
GS11. Take decision with systematic course of actions and/or response
GS12. Planning and organization of work to meet deadlines
GS13. Manage relationships with customers with intent on satisfying its requirements for service delivery
GS14. Follow code of conduct
GS15. Recognize problems and search for solutions

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Identify and Explain Solar PV System Configuration and components	**30**	**20**	**-**	**-**
PC1. discuss the home lighting system configuration and its components	3	-	-	-
PC2. discuss the distributed solar PV system and its components	3	-	-	-
PC3. discuss the charge controller function and its operation	4	-	-	-
PC4. discuss the configuration of charge controller for various solar PV application	4	-	-	-
PC5. discuss the type of charge controller and advantage and disadvantage	4	-	-	-
PC6. discuss the solar PV mini and micro grid system as distributed solar power generation	4	-	-	-
PC7. discuss the small solar pv application for lighting and other application	4	-	-	-
PC8. discuss the configuration and components of standalone Offgrid solar PV system	4	-	-	-
PC9. demonstrate the series and parallel connection of solar module	-	5	-	-
PC10. show how to measure the open circuit voltage (Voc) and short circuit current (Isc) of solar module	-	5	-	-
PC11. show how to measure the Vmp and Imp of solar module	-	5	-	-
PC12. demonstrate different type of solar cell and solar module	-	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4025
NOS Name	Identify and Explain Solar PV System Configuration and identify components
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4026: Perform site assessment and interpret the layout

Description

This unit is about perform site assessment and interpret the layout

Scope

The scope covers the following :

- Introduction of Site survey report
- how to prepare a site survey report

Elements and Performance Criteria

Perform site assessment and interpret the layout

To be competent, the user/individual on the job must be able to:

PC1. introduction of Site survey report and demonstrate the site survey report and discuss its importance

PC2. list out the information collected from the site and include into the report and demonstrate how to prepare a site survey report

PC3. prepare a first-hand layout sketch of site and include into the report

PC4. identify and highlight the major obstacle on site

PC5. discuss about technical specification of existing electrical system (Breaker capacity, spare feeder detail, etc)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's documentation policy

KU4. Obtain authorization from specified field safety officer and supervisor

KU5. Company's reporting structure and Organization culture

KU6. Company's different department and concerned authority

KU7. Relevant Personal protective equipment's required for installation

KU8. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU9. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU10. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

KU11. Know how of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role
GS2. Read English and/or vernacular language
GS3. Read and understand manuals, health and safety instructions, memos, other company documents
GS4. Ability to read from different sources- books screens in machines and signage
GS5. Understand the various color codes, as per standard electrical, mechanica
GS6. Express statements or information clearly so that others can hear and understand
GS7. Participate in and understand the main points of simple discussions
GS8. Respond appropriately to any queries
GS9. Communicate with supervisor
GS10. Follow organization rule-based decision making process

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform site assessment and interpret the layout	**35**	**15**	**-**	**-**
PC1. introduction of Site survey report and demonstrate the site survey report and discuss its importance	6	7	-	-
PC2. list out the information collected from the site and include into the report and demonstrate how to prepare a site survey report	5	8	-	-
PC3. prepare a first-hand layout sketch of site and include into the report	8	-	-	-
PC4. identify and highlight the major obstacle on site	8	-	-	-
PC5. discuss about technical specification of existing electrical system (Breaker capacity, spare feeder detail, etc)	8	-	-	-
NOS Total	**35**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4026
NOS Name	Perform site assessment and interpret the layout
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4027: Install Solar PV DC Standalone system (St. Light)

Description

This unit is about install Solar PV DC Standalone system (St. Light)

Scope

The scope covers the following :

- Standalone Solar PV System and its type
- the Installation of St. Light and other similar system
- the schematics and identify functions of the key components

Elements and Performance Criteria

Install Solar PV DC Standalone system (St. Light)

To be competent, the user/individual on the job must be able to:

PC1. discuss the tools required for the installation of Standalone Solar PV System and its components and demonstrate the use of tools for various application

PC2. discuss the schematic of standalone DC system and demonstrate the working of Solar PV DC standalone system

PC3. discuss the Installation of Solar PV Street light

PC4. discuss the small solar pv application for lighting and other application

PC5. discuss the configuration and components of standalone solar PV system

PC6. show how to measure the open circuit voltage (Voc) and short circuit current (Isc) of solar module

PC7. show how to measure the Vmp and Imp of solar module

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's documentation policy

KU4. Obtain authorization from specified field safety officer and supervisor

KU5. Company's reporting structure and Organization culture

KU6. Company's different department and concerned authority

KU7. Relevant Personal protective equipment's required for installation

KU8. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU9. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU10. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

KU11. Know how of tools & tackles required to carry out the work.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role

GS2. Read English and/or vernacular language

GS3. Read and understand manuals, health and safety instructions, memos, other company documents

GS4. Ability to read from different sources- books screens in machines and signage

GS5. Understand the various color codes, as per standard electrical, mechanical

GS6. Express statements or information clearly so that others can hear and understand

GS7. Participate in and understand the main points of simple discussions

GS8. Respond appropriately to any queries

GS9. Communicate with supervisor

GS10. Follow organization rule-based decision making process

GS11. Take decision with systematic course of actions and/or response

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Install Solar PV DC Standalone system (St. Light)	**30**	**20**	**-**	**-**
PC1. discuss the tools required for the installation of Standalone Solar PV System and its components and demonstrate the use of tools for various application	5	5	-	-
PC2. discuss the schematic of standalone DC system and demonstrate the working of Solar PV DC standalone system	5	5	-	-
PC3. discuss the Installation of Solar PV Street light	6	-	-	-
PC4. discuss the small solar pv application for lighting and other application	7	-	-	-
PC5. discuss the configuration and components of standalone solar PV system	7	-	-	-
PC6. show how to measure the open circuit voltage (Voc) and short circuit current (Isc) of solar module	-	5	-	-
PC7. show how to measure the Vmp and Imp of solar module	-	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4027
NOS Name	Install Solar PV DC Standalone system (St. Light)
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4028: Install Solar Off Grid PV System

Description

This unit is about install Solar Off Grid PV System

Scope

The scope covers the following :

- Solar PV Off Grid System types and its component
- electrical interconnection and assembly of different components
- schematics of system

Elements and Performance Criteria

Install Solar Off Grid PV System

To be competent, the user/individual on the job must be able to:

PC1. discuss the tools required for assembly and interconnection of Solar off grid system and its components

PC2. discuss the MPPT and PWM charge controller and its application in Off Grid solar PV system

PC3. discuss the electrical interconnection and assembly of charge controller, battery and solar panel and demonstrate the various function of Solar charge controller

PC4. discuss the series and parallel connection of battery and demonstrate the series and parallel connection of solar module

PC5. discuss the role of inverter in solar PV Off Grid power system

PC6. discuss the electrical interconnection of Off grid Solar inverter, Battery and Load and demonstrate the various function of Solar off grid Inverter

PC7. demonstrate the series and parallel connection of solar module

PC8. show how to measure the array voltage (Voc) and current (Isc) of the system

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's Customer Support Policy

KU4. Company's documentation policy

KU5. Obtain authorization from specified field safety officer and supervisor

KU6. Company's reporting structure and Organization culture

KU7. Company's different department and concerned authority.

KU8. Relevant Personal protective equipment's required for installation

KU9. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU10. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU11. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role

GS2. Read English and/or vernacular language

GS3. Read and understand manuals, health and safety instructions, memos, other company documents

GS4. Ability to read from different sources- books screens in machines and signage

GS5. Understand the various color codes, as per standard electrical, mechanical

GS6. Express statements or information clearly so that others can hear and understand

GS7. Participate in and understand the main points of simple discussions

GS8. Respond appropriately to any queries

GS9. Communicate with supervisor

GS10. Follow organization rule-based decision making process

GS11. Planning and organization of work to meet deadlines.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Install Solar Off Grid PV System	**25**	**25**	**-**	**-**
PC1. discuss the tools required for assembly and interconnection of Solar off grid system and its components	3	-	-	-
PC2. discuss the MPPT and PWM charge controller and its application in Off Grid solar PV system	3	-	-	-
PC3. discuss the electrical interconnection and assembly of charge controller, battery and solar panel and demonstrate the various function of Solar charge controller	4	5	-	-
PC4. discuss the series and parallel connection of battery and demonstrate the series and parallel connection of solar module	5	5	-	-
PC5. discuss the role of inverter in solar PV Off Grid power system	5	-	-	-
PC6. discuss the electrical interconnection of Off grid Solar inverter, Battery and Load and demonstrate the various function of Solar off grid Inverter	5	5	-	-
PC7. demonstrate the series and parallel connection of solar module	-	5	-	-
PC8. show how to measure the array voltage (Voc) and current (Isc) of the system	-	5	-	-
NOS Total	**25**	**25**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4028
NOS Name	Install Solar Off Grid PV System
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4029: Install Small solar rooftop grid tied plant for residential home

Description

This unit is about install Small solar rooftop grid tied plant for residential home

Scope

The scope covers the following :

- Solar PV residential System its component
- electrical interconnection and assembly of different components
- the schematics of system

Elements and Performance Criteria

Installation of Small solar rooftop grid tied plant for residential home

To be competent, the user/individual on the job must be able to:

PC1. discuss the grid tied inverter and its application and demonstrate the working of solar inverter

PC2. explain how to do balance of system in solar PV Grid tied system

PC3. explain the Junction box, Protection switch and other safety components required in solar PV system and demonstrate the junction box and inbuilt safety components

PC4. discuss the metering arrangement in Solar PV grid tied system

PC5. discuss the various mounting structure types and its installation procedure

PC6. discuss how to maintain the plant and demonstrate how to maintain the plant

PC7. demonstrate the earthing and Lighting Arrester device

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's Customer Support Policy

KU4. Company's documentation policy

KU5. Obtain authorization from specified field safety officer and supervisor

KU6. Company's reporting structure and Organization culture

KU7. Company's different department and concerned authority

KU8. Relevant Personal protective equipment's required for installation

KU9. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU10. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU11. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

KU12. Knowhow of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role
GS2. Read English and/or vernacular language
GS3. Read and understand manuals, health and safety instructions, memos, other company documents
GS4. Ability to read from different sources- books screens in machines and signage
GS5. Understand the various color codes, as per standard electrical, mechanical
GS6. Express statements or information clearly so that others can hear and understand
GS7. Participate in and understand the main points of simple discussions
GS8. Respond appropriately to any queries
GS9. Follow organization rule-based decision making process
GS10. Take decision with systematic course of actions and/or response
GS11. Planning and organization of work to meet deadlines.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Installation of Small solar rooftop grid tied plant for residential home	**30**	**20**	**-**	**-**
PC1. discuss the grid tied inverter and its application and demonstrate the working of solar inverter	3	5	-	-
PC2. explain how to do balance of system in solar PV Grid tied system	5	-	-	-
PC3. explain the Junction box, Protection switch and other safety components required in solar PV system and demonstrate the junction box and inbuilt safety components	3	5	-	-
PC4. discuss the metering arrangement in Solar PV grid tied system	7	-	-	-
PC5. discuss the various mounting structure types and its installation procedure	7	-	-	-
PC6. discuss how to maintain the plant and demonstrate how to maintain the plant	5	5	-	-
PC7. demonstrate the earthing and Lighting Arrester device	-	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4029
NOS Name	Install Small solar rooftop grid tied plant for residential home
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4030: Introduce tools and tackles used for Solar Installation

Description

This unit is about introduce tools and tackles used for Solar Installation

Scope

The scope covers the following :

- role of tools in system installation
- various application of tools
- know-how of tools used in solar Installation

Elements and Performance Criteria

Introduction of tools and tackles used for Solar Installation

To be competent, the user/individual on the job must be able to:

PC1. discuss the role and requirement of tools in solar PV installation

PC2. discuss the various tools types and demonstrate the various tools used in solar pv installation

PC3. discuss the specific function and application of various tools and demonstrate the working of tools

PC4. describe the tools used for site survey, load analysis and demonstrate the site survey tools

PC5. describe the Electrical, mechanical and various testing tools

PC6. discuss how to use the tools

PC7. demonstrate the risk associated with tools

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's Customer Support Policy

KU4. Company's documentation policy

KU5. Obtain authorization from specified field safety officer and supervisor

KU6. Company's reporting structure and Organization culture

KU7. Company's different department and concerned authority

KU8. Relevant Personal protective equipment's required for installation

KU9. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU10. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU11. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plant

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Read English and/or vernacular language
GS2. Read and understand manuals, health and safety instructions, memos, other company documents
GS3. Ability to read from different sources- books screens in machines and signage
GS4. Understand the various color codes, as per standard electrical, mechanical
GS5. Express statements or information clearly so that others can hear and understand
GS6. Participate in and understand the main points of simple discussions
GS7. Respond appropriately to any queries
GS8. Follow organization rule-based decision making process
GS9. Take decision with systematic course of actions and/or response
GS10. Planning and organization of work to meet deadlines

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction of tools and tackles used for Solar Installation	**30**	**20**	**-**	**-**
PC1. discuss the role and requirement of tools in solar PV installation	5	-	-	-
PC2. discuss the various tools types and demonstrate the various tools used in solar pv installation	5	5	-	-
PC3. discuss the specific function and application of various tools and demonstrate the working of tools	5	5	-	-
PC4. describe the tools used for site survey, load analysis and demonstrate the site survey tools	5	5	-	-
PC5. describe the Electrical, mechanical and various testing tools	5	-	-	-
PC6. discuss how to use the tools	5	-	-	-
PC7. demonstrate the risk associated with tools	-	5	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4030
NOS Name	Introduce tools and tackles used for Solar Installation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

SGJ/N4031: Maintain Health and safety measures for installing and operating solar DC system

Description

This unit is about maintain Health and safety measures for installing and operating solar DC system

Scope

The scope covers the following :

- understand the requirements for safe work area
- Identify the hazards associated with photovoltaic installations

Elements and Performance Criteria

Maintain Health and safety measures for installing and operating solar DC system

To be competent, the user/individual on the job must be able to:

PC1. explain the requirements for safe work area

PC2. explain the importance of administering first aid and demonstrate how to administer first aid

PC3. identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work

PC4. identify the hazards associated with photovoltaic installations

PC5. identify work safety procedures and instructions for working at height

PC6. explain the importance of Occupational health & Safety standards and regulations for installation of Solar PV system

PC7. discuss about incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Company's Installation Policy

KU2. Company's work safety policy

KU3. Company's Customer Support Policy

KU4. Company's documentation policy

KU5. Obtain authorization from specified field safety officer and supervisor

KU6. Company's reporting structure and Organization culture

KU7. Company's different department and concerned authority

KU8. Relevant Personal protective equipment's required for installation

KU9. Relevant standards and regulations for installation of Solar Photovoltaic Power Plant in India

KU10. Occupational health and safety (OHS) standards for installation of Solar Photovoltaic Power Plant

KU11. Risk identification and mitigation procedure for safe installation of Solar Photovoltaic Power Plan

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Fill up documentation applicable to one's role
GS2. Read English and/or vernacular language
GS3. Read and understand manuals, health and safety instructions, memos, other company documents.
GS4. Ability to read from different sources- books screens in machines and signage
GS5. Understand the various color codes, as per standard electrical, mechanical
GS6. Express statements or information clearly so that others can hear and understand
GS7. Participate in and understand the main points of simple discussions.
GS8. Respond appropriately to any queries
GS9. Follow organization rule-based decision making process.
GS10. Take decision with systematic course of actions and/or response
GS11. Planning and organization of work to meet deadlines

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Maintain Health and safety measures for installing and operating solar DC system	**29**	**21**	**-**	**-**
PC1. explain the requirements for safe work area	3	-	-	-
PC2. explain the importance of administering first aid and demonstrate how to administer first aid	3	7	-	-
PC3. identify the personal protective equipment used for the specific purpose and demonstrate the usage of personal protective equipment for ensuring safety during installation and O&M work	3	7	-	-
PC4. identify the hazards associated with photovoltaic installations	5	-	-	-
PC5. identify work safety procedures and instructions for working at height	6	-	-	-
PC6. explain the importance of Occupational health & Safety standards and regulations for installation of Solar PV system	6	-	-	-
PC7. discuss about incorporate good housekeeping practices and infection control guidelines and demonstrate good housekeeping and infection control & prevention practices	3	7	-	-
NOS Total	**29**	**21**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4031
NOS Name	Maintain Health and safety measures for installing and operating solar DC system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4023.Introduce Solar PV Technology	26	24	0	0	50	10
SGJ/N4024.Perform DC electrical work of Solar Module components	30	20	0	0	50	10
SGJ/N4025.Identify and Explain Solar PV System Configuration and identify components	30	20	0	0	50	10
SGJ/N4026.Perform site assessment and interpret the layout	35	15	0	0	50	10
SGJ/N4027.Install Solar PV DC Standalone system (St. Light)	30	20	0	0	50	10
SGJ/N4028.Install Solar Off Grid PV System	25	25	0	0	50	10
SGJ/N4029.Install Small solar rooftop grid tied plant for residential home	30	20	0	0	50	10
SGJ/N4030.Introduce tools and tackles used for Solar Installation	30	20	0	0	50	10
SGJ/N4031.Maintain Health and safety measures for installing and operating solar DC system	29	21	0	0	50	10

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	**285**	**215**	**-**	**-**	**500**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.